

What the present synthesis adds to existing reviews of machine learning for geological CO₂ storage

Feature coding is based on verified full text. Silence is unclear, not no. The final row is author synthesis.

	Search transparency	Corpus verification	Decision framing	Method evolution	Physical diagnostics	Uncertainty calibration	Transfer testing	Field validation	MRV auditability	Open traceability	Missingness aware coding	Minimum validation packages
Li et al. (2023)	?	?	P	Y	P	P	?	P	?	?	?	?
Mao et al. (2024)	P	?	Y	P	P	P	P	P	?	?	?	P
Lin et al. (2025)	P	P	P	Y	P	P	P	P	P	?	?	P
Marvin et al. (2025)	P	?	P	Y	P	P	P	P	P	?	?	P
Watheq J. Al-Mudhafar (2026)	?	?	Y	Y	Y	P	P	P	P	?	?	P
de la Cruz-Azuara et al. (2026)	Y	P	P	Y	P	P	Y	P	P	Y	?	?
Chunkai Wang et al. (2026)	?	?	Y	P	Y	P	Y	P	Y	?	?	P
Wenkai Wei et al. (2026)	Y	P	Y	Y	Y	Y	Y	P	P	?	?	P
Yan et al. (2021)	P	?	P	P	P	P	?	P	?	?	?	?
Li et al. (2024)	P	?	P	P	P	P	?	P	?	?	?	?
Yanwei Wang (2024)	NA	NA	P	Y	P	?	?	?	?	?	?	?
Aminaho et al. (2026)	P	?	P	P	P	P	?	P	P	?	?	P
Roberts and Stalker (2020)	P	?	NA	NA	Y	P	NA	Y	P	?	?	NA
Bashir et al. (2024)	P	?	NA	NA	Y	P	NA	Y	P	?	?	NA
Present synthesis	P	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y

Yes Partial Unclear No Not applicable